# Automatic Air Quality Robot

Andy Liu & Jeffrey Lee

# Problem: Indoor Air Quality

- Allergic Reactions
- Respiratory conditions / illnesses
- Overall discomfort
- Etc.



# Solutions & Limitations:

- Air Conditioning System (to address air circulation and discomfort from humid and hot air indoors)
  - Costly, consumes large amounts of energy
- Air Purifiers (to address concentration of air particles in the air)
  - Costly, limited range



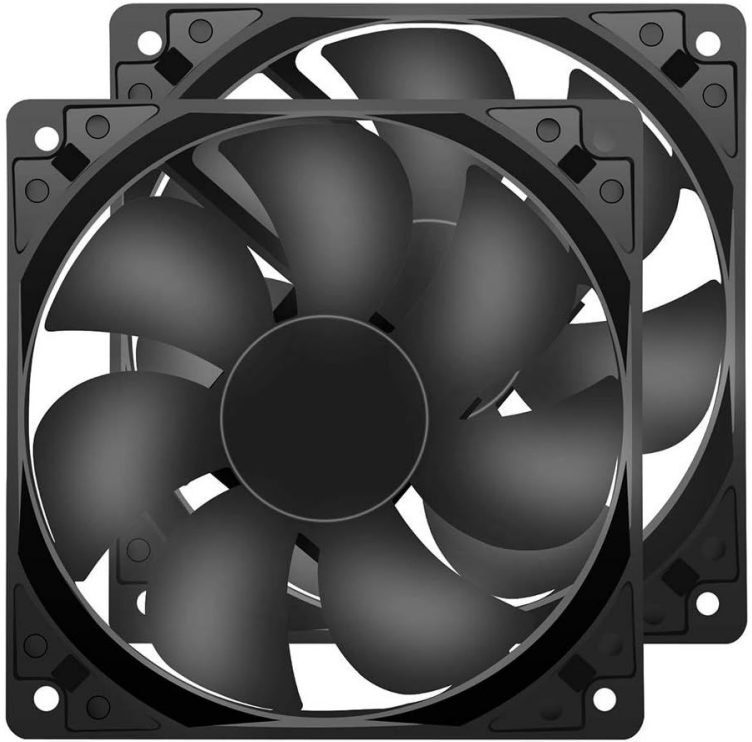
# Proposal:

- Most cases: want to cool/dehumidify/purify a specific location, especially locations exhibiting high temperatures, humidity, and concentrations of particulate matter
- Emphasizing portability for a cost and energy effective way to focus/target spots where improved air circulation/quality is desired
- Similar products:
  - Automated AC systems (eg. smart homes)
  - Automated Home Cleaning Robots (eg. Roomba)
- Idea: respond to temperature, humidity, and air quality to target specific locations for air circulation and purification

# Hardware Components:

- Robot Car Chassis (1)
- DC Motors (4)
- Wheels (4)
- L298N Motor Driver (1)
- PMS5003 (& Adapter Kit) (1)
  - USB-Serial Converter (1)
- Lithium Ion Batteries (2)
- Relay Modules (2)
- DHT11 Module (1)
- Ultrasonic Sensor (1)
- Fan (1)
- Negative Ion Generator Module (1)






负离子发生器
DC5V<1W
红线接"+"黑线接"-"

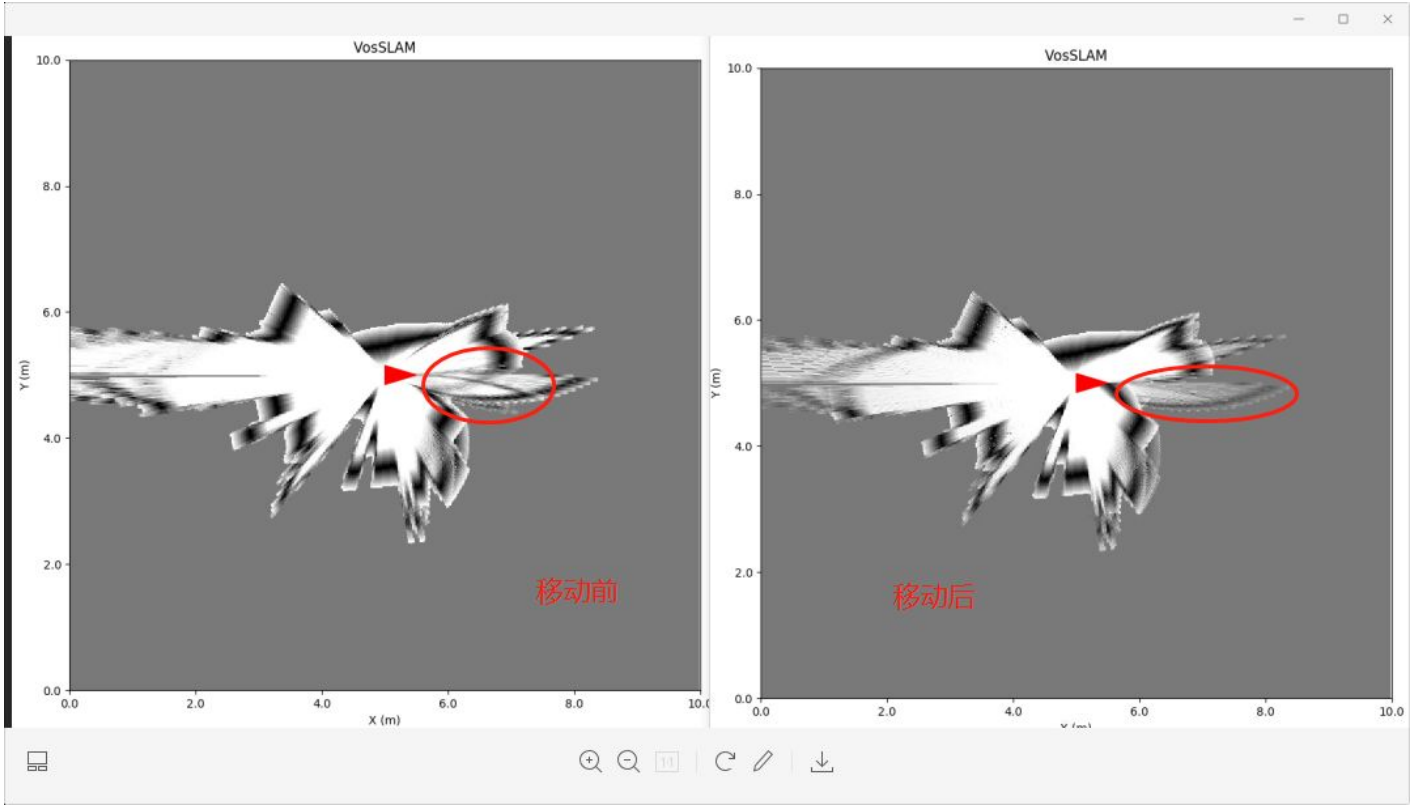
# Air Purification/Filtering: Negative Ion Generator Module



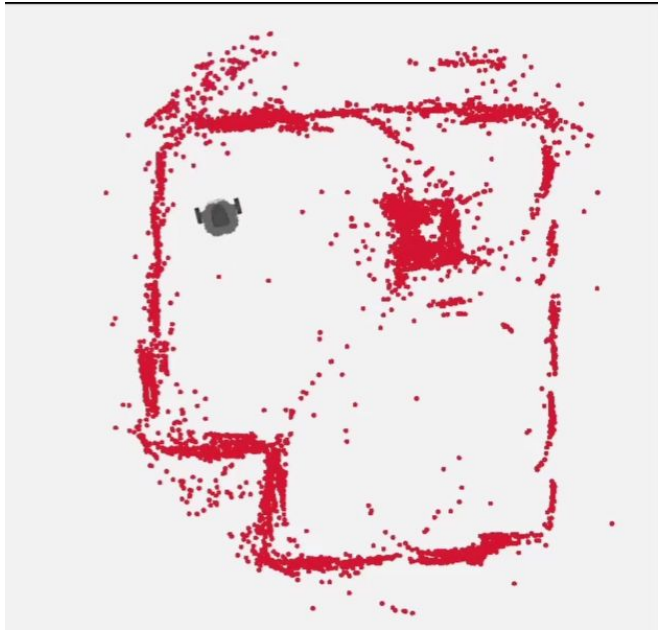
# Software Components

- Simultaneous Localization and Mapping (SLAM)
  - BreezySLAM
    - Imitated lidar sensor with ultrasonic
  - Odometry function to measure distance, inaccurate
  - Car autonomously avoids obstacles
- Main code used Python
- Sensor code used C with WiringPi library
- ctypes library to wrap C code for Python use
- Visualization (for mapping) with Matplotlib

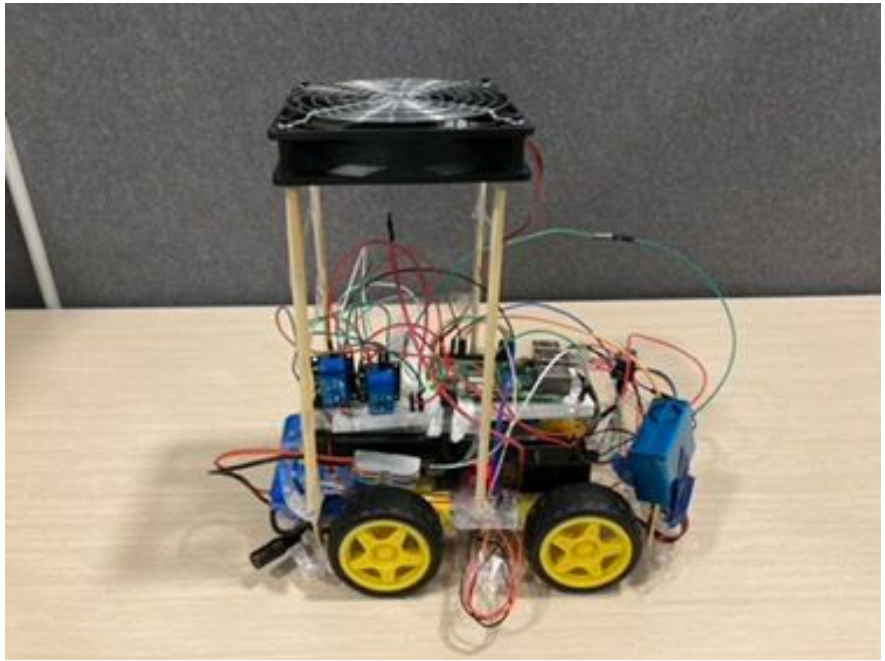
# BreezySLAM

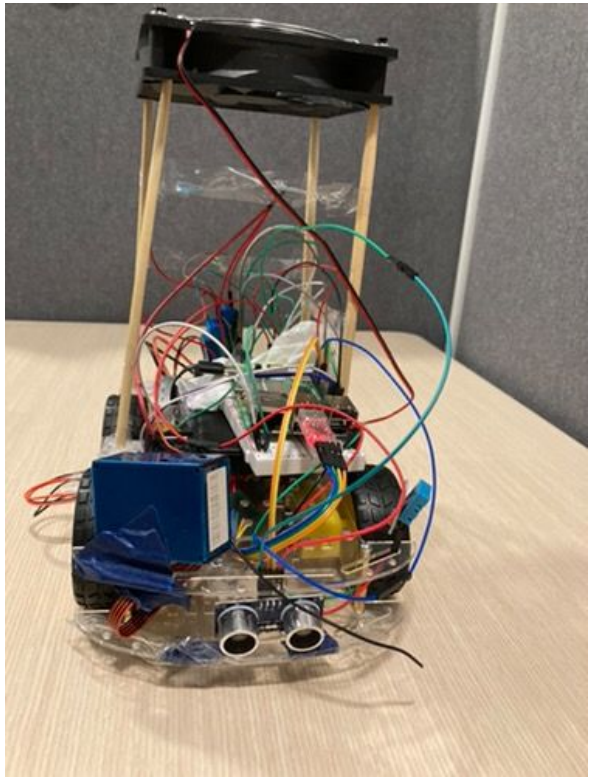

VosSLAM
10.0
8.0
6.0
4.0
2.0
0.0
Y (m)
0.0
2.0
4.0
6.0
8.0
10.0
X (m)
移动前
VosSLAM
移动后


ELEGOO
Object



# Demo





# Demo



# Project Overview

- Workload Distribution:
  - Andy: Software (mapping & navigation algorithm, data processing, visualization)
  - Jeffrey: Hardware (circuitry, assembly, sensor/actuator drivers, sensor-actuator interaction)
  - Both: research, testing, documentation, etc.
- Challenges & Limitations:
  - Wanted it to be minimalistic (only RPi5, bare essentials for robot)
  - Due to time and cost constraints: inconsistent/inaccurate mapping
  - More research needed to be done ahead of time (especially for assembling the components together)
  - Need more time to tune mapping and navigation algorithm

# Next Steps / Future Implementation

- Implement Lidar sensor for faster, more accurate, and greater range for mapping
- Implement accelerometer & gyroscope for more accurate and measurable localization
- Fine tune the SLAM algorithm (more time for testing and measurements) and implement more complex navigation algorithms
- Optimize scheduling and efficiency of the robot
- Implement more informative and interpretable visuals (given more accurate mapping)
- Test and measure ability to improve local indoor air quality